

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

BRAIN TUMOR DATA ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

Framework for Data Visual Analytics

SWETHA VEERAMANI

241501261

ABSTRACT

This mini project, carried out for the subject Framework of Data Visualization and Analytics (FDVA), analyses a Brain Tumor Prediction dataset using Python, Microsoft Power BI, and Tableau. The dataset contains patient details such as age, gender, tumor location, tumor size, treatment type, survival rate, diabetes, family history, and genetic risk. Python with Pandas was used for data handling and Matplotlib was used to create visualizations, including bar charts, line charts, scatter plots, and histograms. Interactive dashboards were developed in Power BI and Tableau using charts, maps, filters, and KPI cards to explore patient information, treatment distribution, survival patterns, and risk factors. The dashboards provide a clear view of tumor-related characteristics and comparisons across different countries and patient groups. This project demonstrates how data preprocessing, visualization, and interactive dashboards can transform medical data into meaningful insights and support better understanding of brain tumor-related patterns.

1. Data Preprocessing

The Brain Tumor Prediction dataset contains **250,000 rows and 21 columns**. It was processed using Python and Pandas to prepare the data for visualization in Matplotlib, Power BI, and Tableau.

1.1 Loading and inspecting the data

```
import pandas as pd
df = pd.read_csv("Brain_Tumor_Prediction_Dataset.csv")
print(df.shape)
print(df.info())
```

The dataset contains patient details such as age, gender, country, tumor size, tumor location, genetic risk, treatment, survival rate, and brain tumor presence.

1.2 Missing values and duplicates

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
```

The dataset contains **62,748 missing values in the** Treatment_Received column. No duplicate rows were found

1.3 Fixing data types and filling missing values

```
df["Treatment_Received"] = df["Treatment_Received"].fillna("Unknown")
print(df.isnull().sum())
```

Missing treatment values were replaced with "Unknown" to preserve the patient records without assuming a treatment was received.

1.4 Standardising text values

```
cols = df.select_dtypes(include="object").columns
for col in cols:
    df[col] = df[col].astype(str).str.strip().str.upper()
```

Text values were standardized by removing extra spaces and converting them to uppercase. This helps maintain consistency in categorical variables such as gender, tumor location, treatment, and symptom severity.

1.5 Checking Data Types

```
print(df.dtypes)
```

The numerical columns, including Age, Tumor_Size, Genetic_Risk, and Survival_Rate(%), were checked to ensure they were suitable for analysis. Categorical columns were retained for grouping and visualization.

1.6 Exporting the Cleaned Dataset

```
df.to_csv("Cleaned_Brain_Tumor_Dataset.csv", index=False)
print(df.shape)
```

The cleaned dataset contains **250,000 rows and 21 columns**. It was saved as a CSV file and used for creating visualizations in Python, Power BI, and Tableau.

2. Data Visualization Using Python

I. Bar Chart – Gender vs Brain Tumor

Shows the number of male and female patients with and without brain tumors.
Helps compare tumor occurrence across genders.

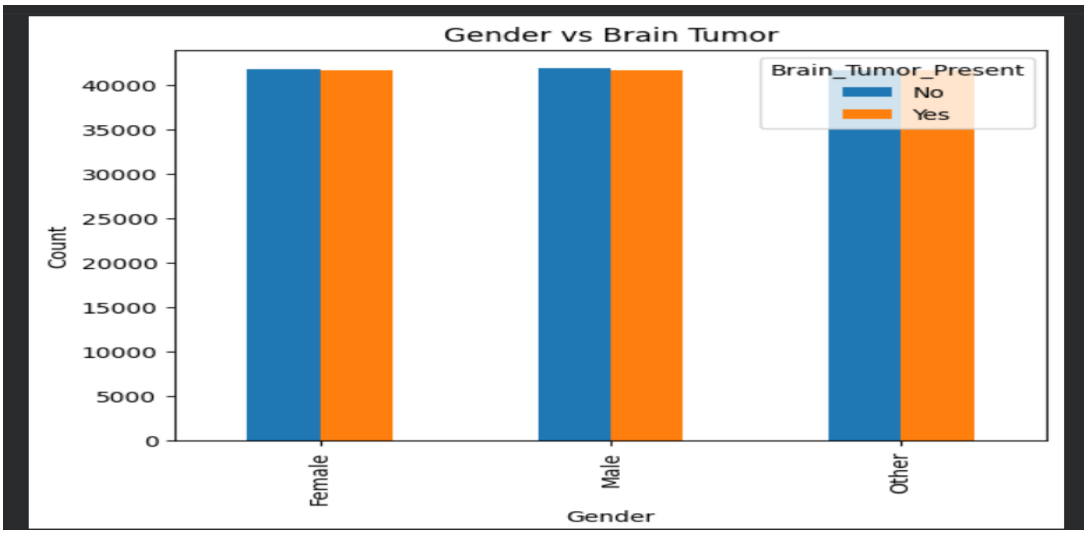


Figure 2.1: Gender wise Brain tumor

Result: Male patients have slightly higher tumor counts than female patients.

II. Line Chart – Age vs Tumor Size

Shows how the average tumor size changes with age.Helps identify trends between age and tumor size.

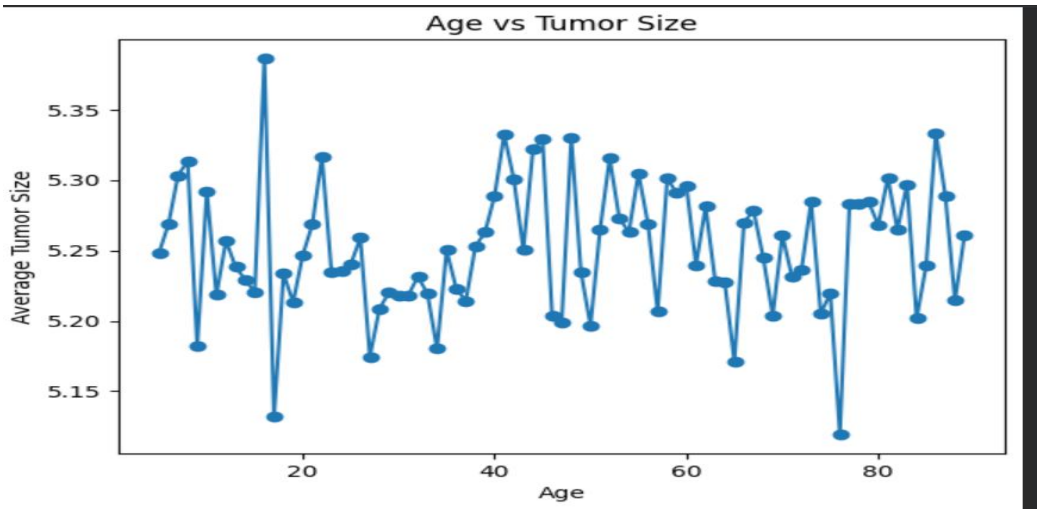


Figure 2.2: Tumor size upon age

Result: Patients aged 50–60 have the highest average tumor size (5.28).

III. Scatter Plot – Age vs Tumor Size

Shows the relationship between patient age and tumor size.Helps identify possible correlation or patterns

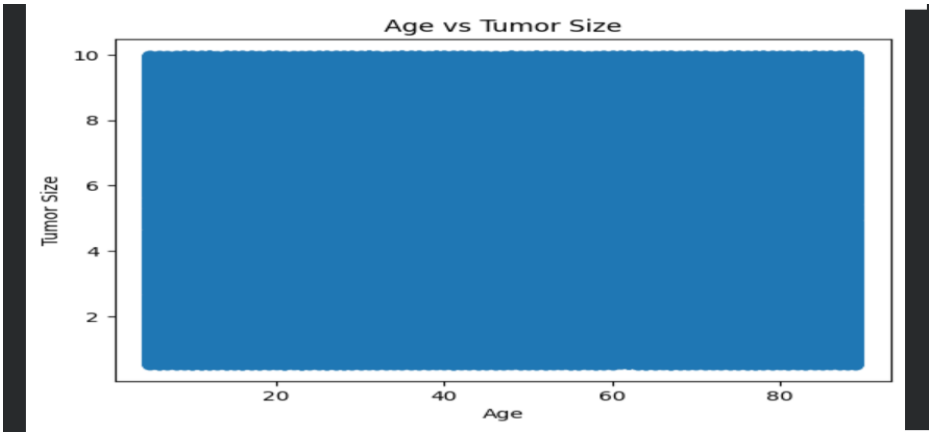


Figure 2.3: Age Vs Tumor size

Result: Tumor size varies with age, with no clear relationship.

IV. Histogram – Age and Tumor Size

Shows the distribution of age and tumor-size values.Helps identify the most common value range

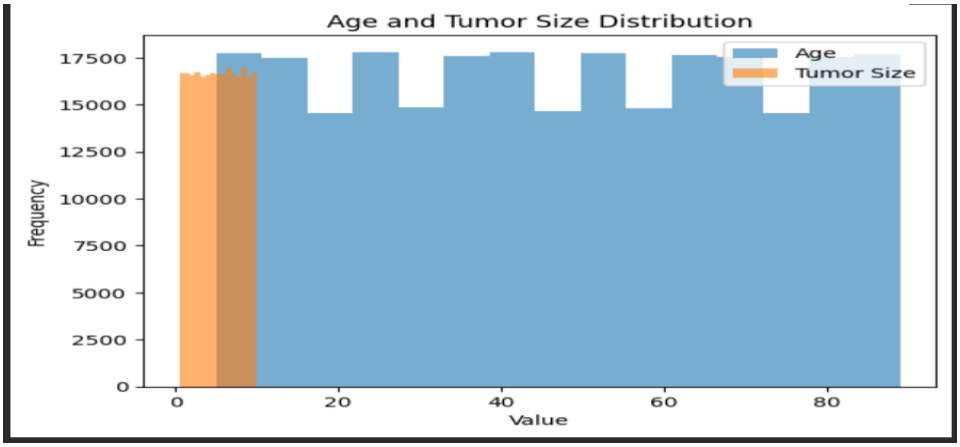


Figure 2.4: Age by Tumore size

Result: Most patients are around 47 years old.

V. Pie Chart – Brain Tumor Present (yes) vs Brain Tumor Present (no)

The pie chart shows the percentage of patients with and without brain tumors. Patients without tumors are slightly higher (50.05%) than patients with tumors (49.95%). Both categories have nearly equal percentages. It provides a clear overview of tumor presence in the dataset.

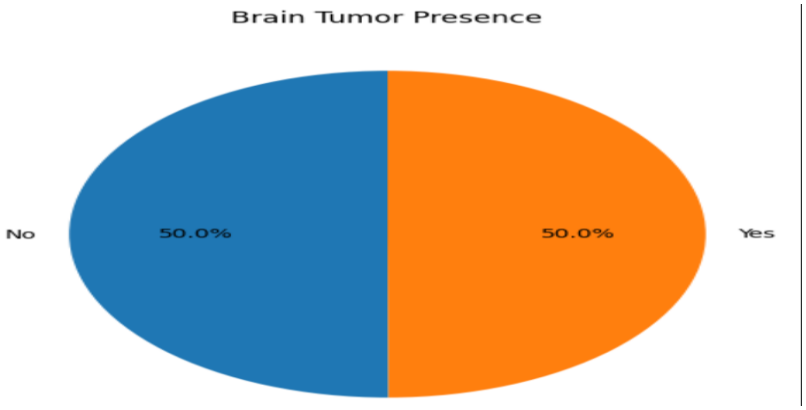


Figure 2.5: Presence and Absence of Brain Tumor

Result: The pie chart shows the percentage of patients with and without brain tumors. Patients without tumors are slightly higher (50.05%) than patients with tumors (49.95%).

VI. Correlation Heatmap -Age vs Tumor Size vs Genetic Risk vs Survival Rate (%) vs Tumor Growth Rate

The correlation heatmap shows the relationship between numerical features such as Age, Tumor Size, Genetic Risk, and Survival Rate using different colors

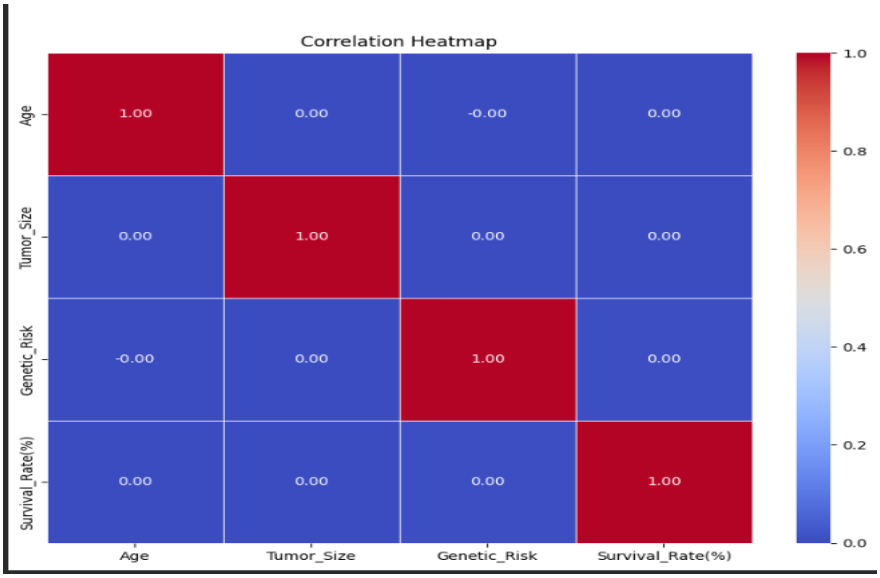


Figure 2.6: Correlation Heatmap

Result: Shows correlations between numerical features.

3. Power BI Dashboard

Power BI was used to build a one-page interactive dashboard from the cleaned brain tumor data.

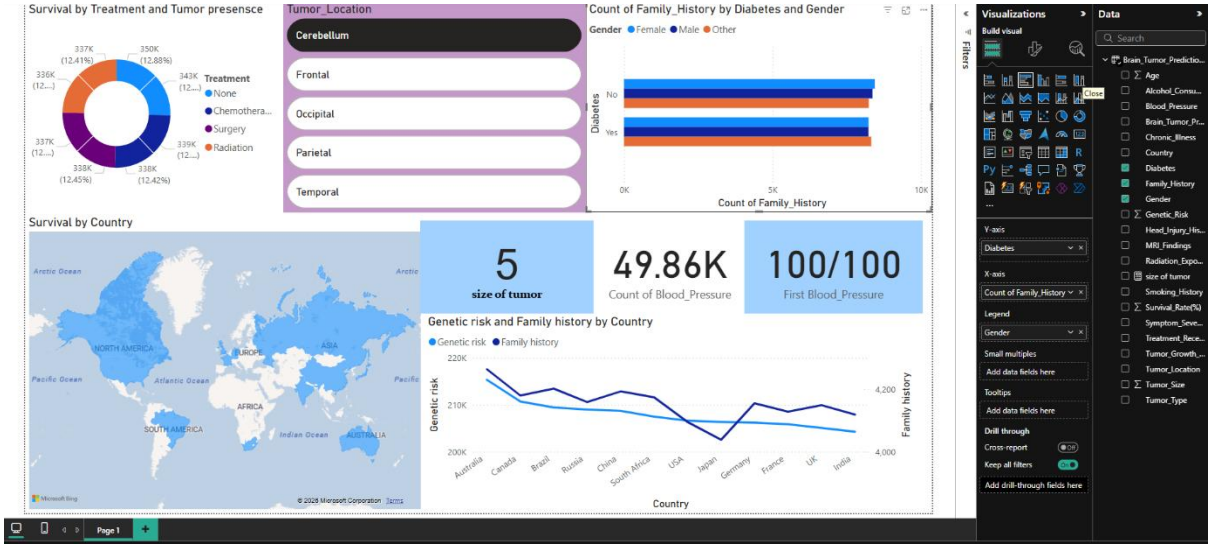


Figure 3.1: Brain Tumor Dashboard in Power BI

- **Survival by Treatment and Tumor Presence – Donut Chart**

  Shows the distribution of patients based on treatment type and tumor presence.

  Helps compare survival-related treatment categories**.**

- **Tumor Location – Slicer**

  Allows users to filter the dashboard by tumor location such as Cerebellum, Frontal, Occipital, Parietal, and Temporal**.**

- **Family History by Diabetes and Gender – Bar Chart**

  Compares family history counts based on diabetes status and gender.

  Helps identify demographic patterns.

- **Survival by Country – Map**

  Displays country-wise distribution of survival-related data.

  Helps identify geographical patterns.

- **Size of Tumor , Count of Blood Pressure , First Blood Pressure– Card**

  Displays the selected tumor size, total blood pressure records, and first blood pressure value as key indicators.

- **Genetic Risk and Family History by Country – Line Chart**

  Compares genetic risk and family history across different countries.

  Helps identify trends and variations between countries.

4. Tableau Dashboard

Tableau was used to build the “Brain Tumor Analysis Dashboard” from the same cleaned data.



Figure 4.1: Brain Tumor Dashboard in Tableau

- **Filters**: Tumor Location, Gender, Tumor Type, and Treatment Received filters, with patient age selection.
- **KPIs:** Tumor Size, Total Blood Pressure Records, and First Blood Pressure Value.
- **Charts**: Survival by treatment and tumor presence, family history by diabetes and gender, survival by country map, and genetic risk by country.
- **Insights**: Tumor counts are nearly equal across genders, Parietal has the highest tumor-location count, and patients without tumors are slightly higher than patients with tumors.